



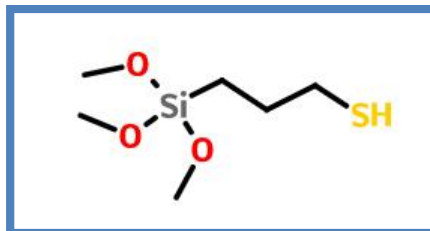
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3-Mercaptopropyltrimethoxysilane

CAS: 4420-74-0



Product Information

LT-590

3-Mercaptopropyltrimethoxysilane

Molecular formula: $C_6H_{16}O_3SSi$

Molecular weight: 196.34

CAS No. : 4420-74-0

Product Features

LT-590 It is an organosilane with an organic sulfhydryl group and a hydrolyzable methoxysilyl difunctional group. This bifunctional structure gives LT-590 special properties: it is able to bond inorganic materials (such as glass, metal, fillers) and organic polymers (such as thermoplastics, thermosets or elastomers) together organically, thus acting as adhesion, crosslinking agents and/or surface modifiers.

It is a colorless transparent liquid with a slight odor of mercaptan, soluble alcohols, ketones, aliphatic or aromatic hydrocarbons.

Physicochemical data

Properties	Numerical value
Appearance	Colorless clear liquid
purity	96.0% or higher
Density 25 ° C	1.057 g/ml
Boiling point 760mmHg	213 ~ 215 °C
Closed cup flash point	76 °C
Refractive index 20 ° C	1.440

Note: The above data is for reference only and cannot be formulated as technical specifications



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Reaction properties

LT-590 In the presence of water, the methoxy group will be hydrolyzed to form a reactive silanol group, and can form a bond with a variety of inorganic substrate surface. The sulfhydryl group of LT-590 with organic properties can react and bond with suitable organic polymers.

Hydrolysis of LT-590 can be catalyzed by an acid reaction.

Suitable inorganic materials: glass, fiberglass, silica, quartz, sand, cristobalite, mica, kaolin, talc, other silicate seasonings, and metals.

Product Application

The LT-590 is an important component in many industrial applications, such as:

- ✧ Glass composites: as an integral or infiltrating component
- ✧ Glass and metal primers
- ✧ Sealants and adhesives: as primers, additives, or chemical modifications to polymers
- ✧ Inorganic fill-type complexes: Pretreat fillers or pigments or act as additives to polymers

The LT-590 also brings important properties to the end product, including:

- ✧ Compression deformation
- ✧ Improve mechanical properties such as bending strength, tensile strength, impact strength, tear strength, and elastic modulus
- ✧ Improve moisture resistance and corrosion resistance
- ✧ Improve electrical properties, such as dielectric constant, volume resistivity
- ✧ Improve thermal resistivity

Improve processing properties such as:

- ✧ Curing time
- ✧ Dispersity
- ✧ Rheological behavior (e.g. reduced viscosity)



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✧ Increase filler load

Product security, handling and storage

Before using the product, you should carefully read the latest product safety technical information and toxicity data of the product as well as the correct storage and use information of the product.

The container should be sealed after opening to prevent water vapor from entering and hydrolyzing.

The effective time of the product is indicated on the label. Beyond this time, it does not mean that the product is no longer usable, and can still be used after the performance test confirms that it is qualified.

This product information as well as all technical advice is based on our current level of experience and knowledge. We accept no liability whatsoever, including for the intellectual property rights of known third parties. In particular, we do not give any guarantee or guarantee in a legal sense about the nature of the products. We reserve the right to make any changes to the product information due to technological progress or further development. The customer carefully tests and inspects the product itself. This description of the nature of the product should also be verified by the Customer through professional testing. References to grades used by other companies are neither recommendations nor exclusions from the availability of similar products.

Hubei New Blue Sky's sole warranty is that the products meet the New Blue Sky sales specifications at the time of shipment.

Your only remedy for breach of Warranty by Hubei New Blue Sky is limited to a refund of the purchase price or replacement of any product that does not comply with the WARRANTY.